

Nesting comparison report

Source: examples/inputs/repeating_batch.dxf DXF: AC1032 Unit: millimeters

Sheet: 1220.0 x 2440.0 mm Parts: 3 unique / 76 total

Metric	symmetry-first	waste-first
Material utilization	23.8%	35.7%
Utilization (gross of kerf)	25.6%	38.3%
Waste	76.2%	64.3%
Sheets used	6	4
Unique sheet layouts	5	4
Sheet replication efficiency	0.20	0.00
Parts placed	76	76
Parts unplaced	0	0
Total cut length	104,219 mm	104,219 mm
Pierce count	112	112
Rapid travel (est.)	26,441 mm	24,965 mm
Direction changes	4832	4832
Min edge clearance	10.0 mm	17.0 mm
Waste skeleton connected	False	True
Skeleton components	7	4
Common-line candidates	0	0
Runtime	0.01 s	1.43 s

Recommendation: waste-first (scores: {'symmetry-first': 0.3, 'waste-first': 0.337})

- waste-first needs 2 fewer sheet(s)
- waste-first achieves 11.9 points higher utilization
- waste-first keeps the waste skeleton connected (easier clamping)

















